Project Manual

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https://drive.google.com/file/d/1dWruv_T5w4rfocttBrBDPkc7lz4Cxeyl/view?usp=sharing

Description

Carbon monoxide (CO) is a poisonous gas that is both odorless and colorless, making it undetectable to human senses. Being exposed to too much of it can cause serious illness, or even death. Because of this, we are creating a system that will use sensors to detect CO levels in an area and create a high or low output, depending on the amount detected. In milestone 1, we created a system to light up an LED if unsafe levels of CO were detected. In milestone 2, we built on top of this, adding a ventilation system. When unsafe levels of CO are detected, the ventilation system turns on. Because higher levels of CO will take longer to bring down, we implemented a way to increase fan speed the longer the ventilation system is active. This means that higher levels of CO will end up resulting in higher fan speeds, and lower levels won't require this. This is our way of creating energy efficiency in our design. In our final stage of MS3, an audible alarm is implemented to create a beeping sound whenever the milestone 1 circuit decides the CO levels are unsafe. The sound will continually beep as long as the trigger is high and only when CO levels are safe again (MS1 indication OFF) then it will be silent.

The block diagram for milestones 1-3 is shown below in figure 1. Here we are using an op amp as a comparator to do our basic decision making: if unsafe levels of CO are detected then turn on the LED and if not, the LED stays off. We have two inputs going into the op amp. The first is the output from the sensor after passing through an RC Low Pass Filter. The second is our reference voltage that we got by using a potentiometer. This reference voltage will equal the voltage of the output when the sensor detects 50 parts per million (ppm) which was a value we sourced from OSHA [1]. The output from the comparator will then be connected to an LED which will turn on when the sensor voltage is greater than the reference voltage then turn off when the sensor voltage is less than the reference voltage.

In milestone 2, we use this same output from the comparator. This high/low voltage is fed to an op amp with capacitive feedback. If there is a high voltage input from the comparator in milestone 1, the integrator will begin to output a ramping voltage. The output of the integrator is connected to a 555-timer used as an astable multivibrator. The astable multivibrator creates a voltage pulse and as the voltage ramp increases, the duty cycle also increases. The duty cycle is the ratio of the time the astable multivibrator is operating to the total time of one cycle. The output of our astable multivibrator is fed to a DC brushless fan which is acting as our ventilation system. The higher the duty cycle, the faster the DC motor in the fan will spin.

Finally in milestone 3, the same comparator output is used again to trigger an alarming system. This signal first goes into an NMOS transistor to decide if the comparator output is truly on or off. Then, the output of the transistor triggers the 1st oscillator circuit that has a 1Hz frequency output that will be the signal for time on and time off of the audio output. This is done by connecting the output of the 1st oscillator circuit to the reset of the 2nd oscillator circuit that has the actual frequency output (2kHz) that the speaker will play the note of. Before being sent to the speaker, the frequency output of the 2nd oscillator circuit is put through first a 1st order passive high pass filter with a cutoff frequency of around 1.6kHz. Then the output of this filter is put through a 2nd order active low pass filter with a cutoff of around 3kHz. These filters are used to prevent extreme ranges of frequencies that aren't the audio signal that we want to be playing.

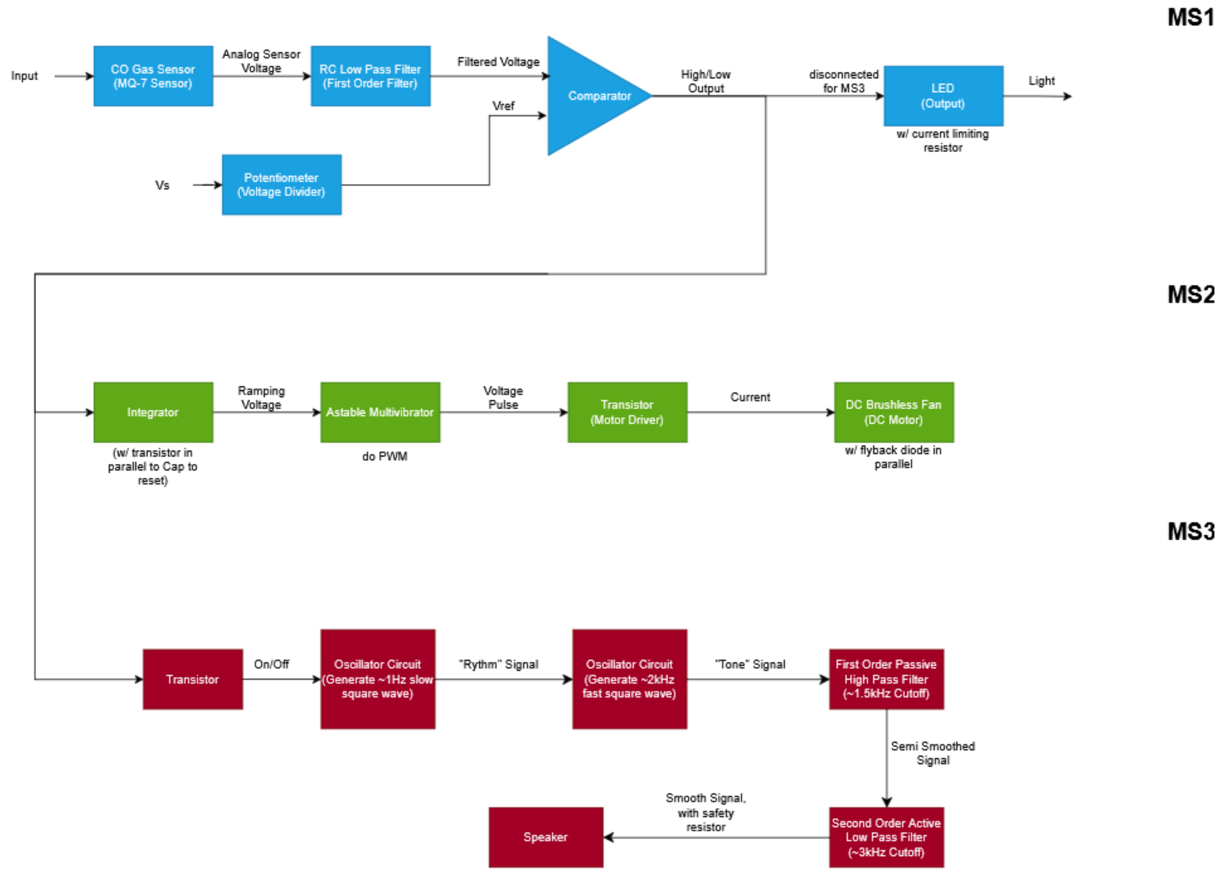


Figure 1: Complete Block Diagram for Milestone 1, 2, and 3

The complete schematic for milestones 1, 2, and 3 is shown below in figure 2. We have two 5-volt voltage sources to power our op amp and provide our input voltages for the op amp. The +5V source is the voltage source to get our reference and sensor voltage. Two 10k potentiometers were used in milestone 1 to set the reference voltage which was left untouched and to simulate the sensor voltage which was done by turning the knob. For example, we can simulate high levels of CO by turning the knob one way to make Rs1 small and turn it the other way making Rs1 large to simulate low levels of. This sensor voltage is then put through the low pass RC filter to filter any noise from the voltage output of the sensor. This is based on the property of a capacitor where it is unable to change voltage instantaneously in cases where the signal frequency is high and only when signal frequency is low can the change be noticeable. To compare the voltages, the reference voltage is connected to the negative input of the op amp, and the filtered sensor voltage is connected to the positive input. Since an ideal op amp has infinite gain, if the sensor voltage is greater than the reference voltage, the output voltage will be saturated at V_{cc+} and if the sensor voltage is less than the reference voltage, the output voltage will be saturated to V_{cc-}. When we have a positive output, the LED will light up and if the output is negative, the LED will be off.

For our integrator design in milestone 2, we use this output (V_{comp}) as the input to our system. As you can see below, this input is then brought to two different op amps. The top op amp is used as a comparator and is used to discharge the capacitor. When our milestone 1 system detects that the CO levels are safe, the output of the comparator will be equal to V_{cc-}(-5V). This low voltage is brought to the comparator and is compared to the reference voltage. Since in this scenario, the voltage reference is greater, and the reference voltage is connected to the positive input, the comparator outputs a high voltage. This output is connected to a NMOS transistor, and when high, connects the output of the integrator to negative input. In an ideal op amp, the voltage at the negative input equals the voltage at the positive input, so in this case, the negative input is basically ground. In conclusion, the capacitor is discharged when the NMOS transistor is active. In addition to this, V_{comp} is also brought to the lower op amp. This op amp is a non-inverting amplifier that acts as an inverter. So, when V_{comp} is a high voltage, indicating unsafe CO levels, it will be a negative value with the same magnitude after this inverter. This output is connected to the negative input of our op amp with capacitive feedback. Using capacitive feedback, we can take advantage of the relationship between the current and voltage of a capacitor ($i_c = C\frac{dV_c}{dt}$) to create a component that integrates the input. This integration is what allows us to get our ramping voltage, which is then brought to the astable multivibrator. This is a 555 circuit that continuously charges and discharges a capacitor through resistors, producing repeated square waves whose duty cycle increases at the ramping voltage increases.

In milestone 3 we implemented an alarm system. At the very beginning of this system is a transistor that we use as an on/off detector. As stated above, we are using the output of the milestone 1 comparator (V_{comp}) to "tell" our system if there are unsafe levels of CO or not. However, the output of the comparator can have some noise so, we implemented this transistor to create a clean, on/off logic signal. Since we want the alarm to go off and alert everyone when the CO levels are dangerous, when the comparator outputs a high voltage, the NMOS transistor is activated, connecting V_{cc+} to the output of this system (V_{trans}). This is the initial step of our milestone 3 system. Next, the output of this transistor system is brought to the active low reset input of our first 555 timer in this milestone, which acts as an oscillator circuit with a very low frequency output. We want the alarm to create a beeping noise, rather than a long continuous noise, so we are using this output to do so. This output will set the rhythm of the alarm by connecting this signal to the reset input of our next 555 timer. This next timer will be used for our 2kHz oscillator circuit. From this, we will get an output which will eventually be brought to the speaker. Considering the input from the first oscillator circuit, we will have a repeating cycle of on then off. Before reaching the speaker, this signal passes through a system of two filters in order to prevent any extreme ranges of frequency that we don't want to be outputted by the speaker. The first filter in the system is a first order passive high pass filter that attenuates any frequencies lower than the cutoff frequency. We achieved this by creating a simple RC circuit and using the voltage of the resistor as our output. Next, this output is brought to the second order active low pass filter which attenuates any frequencies greater than the cutoff frequency of this filter. Lastly, this output is brought to our speaker, causing our alarm to sound.

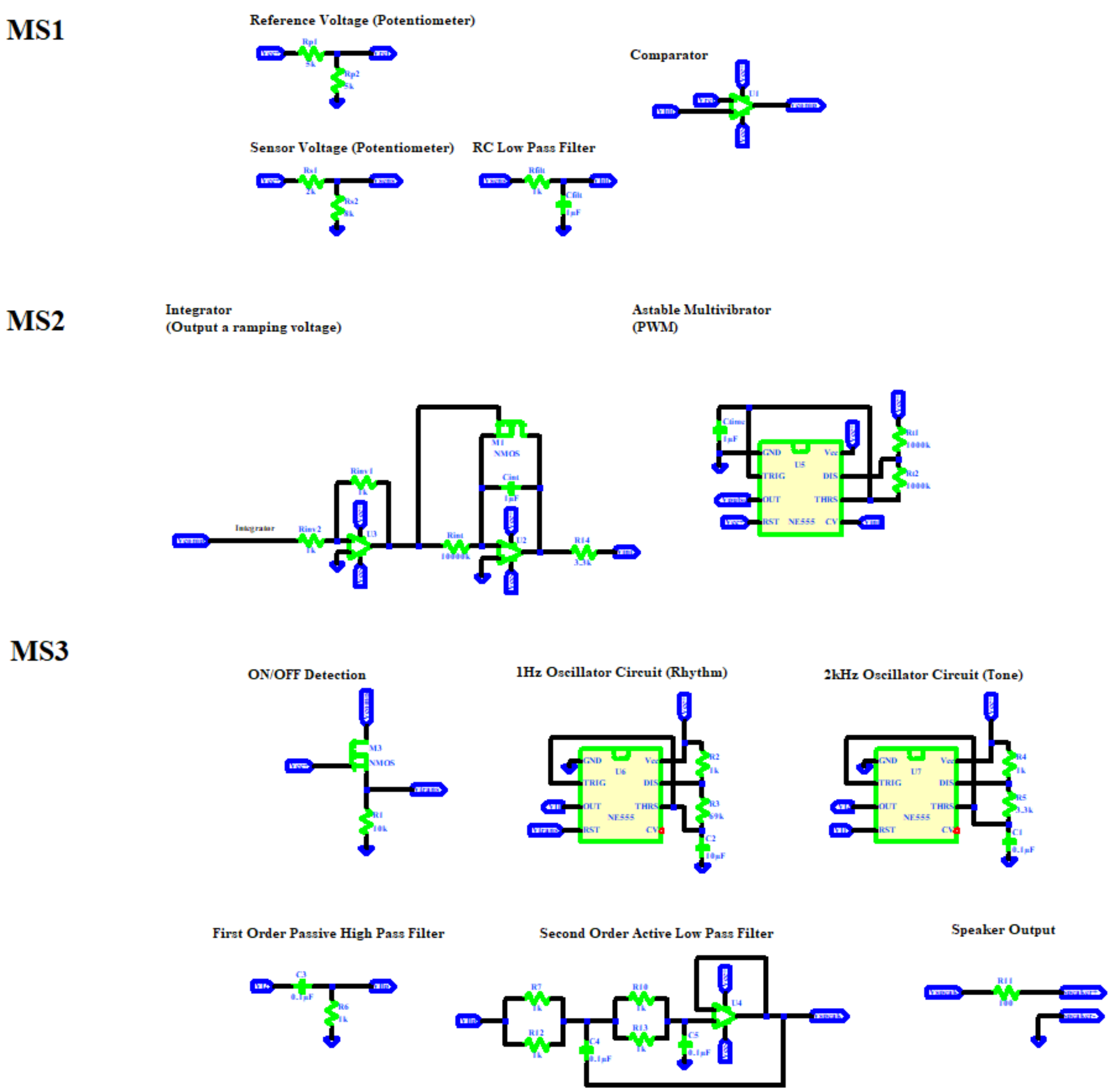


Figure 2: Complete Schematic for Milestone 1, 2, and 3

A pre-existing application of a CO sensor is the household CO sensor by first alert. The sensor's output is typically a low-level analog voltage. Before getting to the decision-making stage, the circuit includes an op amp to amplify the signal to a readable voltage level and a low-pass RC filter. After this, it used a microcontroller to decide: ~50 ppm an LED indicator appears, ~150 ppm causes an alarm, ~400 ppm causes an immediate alarm indicating critical danger [3]. Since we are tasked to design our labs as entirely analog systems, we decided to use an op amp to do the decision making.

Operation and Design

Input, Potentiometer for Sensor Voltage

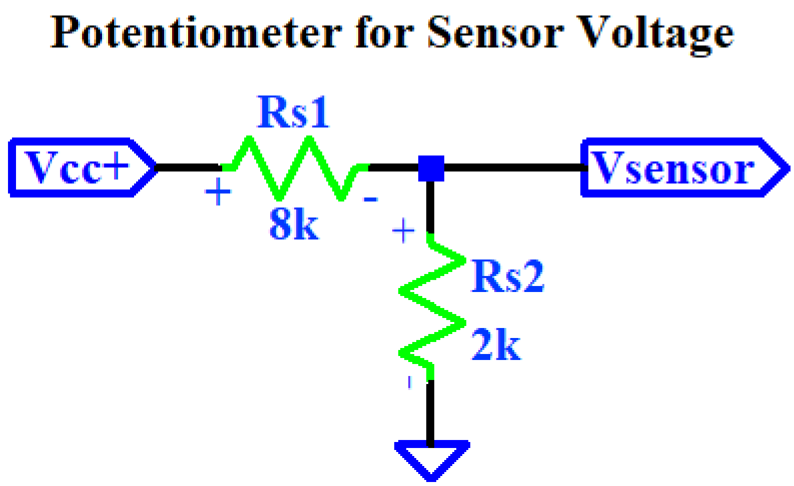


Figure 3: Input Block Schematic

Using the voltage divider equation:

$$V_{sensor} = V_{cc+}\left(\frac{R_2}{R_1 + R_2}\right) \tag{1}$$

Applying this formula to our specific input block gives us the equations relating to the input and output:

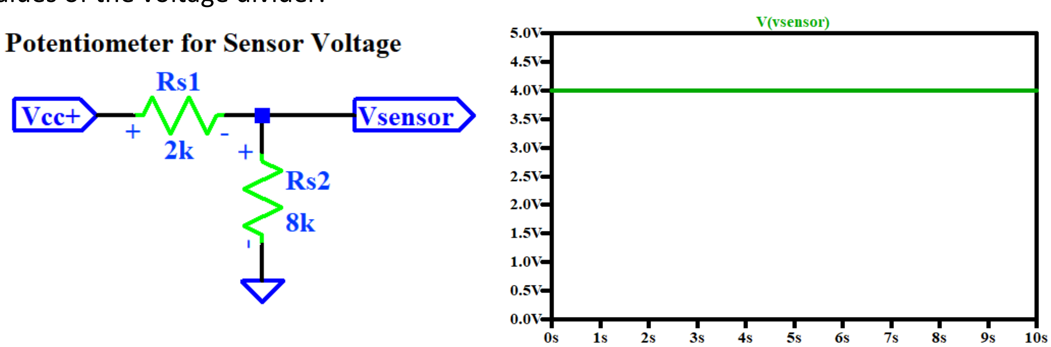
$$V_{sensor} = 5\left(\frac{2k}{10k}\right) \tag{2}$$

Since we vary this potentiometer to simulate different outputs from the CO sensor, here is a more general equation:

$$V_{sensor} = 5\left(\frac{R_{s2}}{R_{s2} + R_{s1}}\right) \tag{3}$$

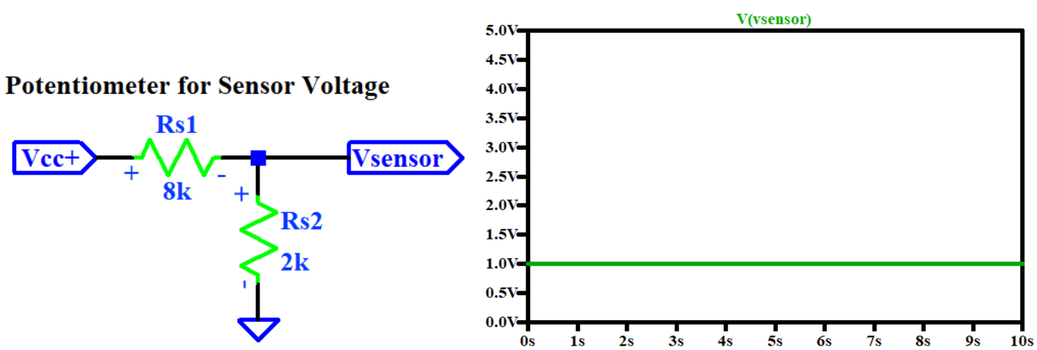
For our sensor voltage, since we are using it for a comparator, we simply simulate for results where the sensor voltage is greater than and when it’s less than the reference voltage of 2.5V.

And so, to get a sensor voltage greater than 2.5V, we turn the knob to the right to change the resistance values of the voltage divider:



Rs1 and Rs2 have a relationship where Rs1 < Rs2 now, the sensor voltage is greater than the reference voltage and in this case is set to 4V.

Now turning the knob to the left:



Rs1 and Rs2 now reverse their relationship to Rs1 > Rs2 which results in the sensor voltage becoming less than 2.5V and in this case 1V.

MS1 Building Block, Comparator

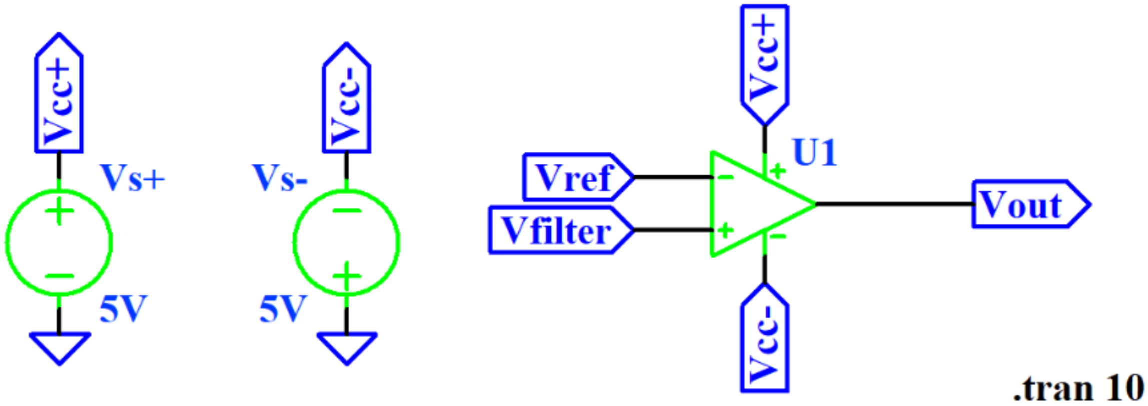


Figure 5: Op amp comparator schematic

We have the reference voltage connected to the negative input of the op amp and the filtered voltage from the sensor connected to the positive output of the op amp.

Because of this, the behavior of the op amp as a comparator is:

$$V_{ref} > V_{filter} \rightarrow V_{out} = V_{cc-} = -5V \quad (4)$$

$$V_{filter} > V_{ref} \rightarrow V_{out} = V_{cc+} = +5V \quad (5)$$

We chose this specific op amp design to allow our circuit to do very basic decision making. This design lets us set a reference voltage and turn on an LED if the other input is greater than this reference value. The specific op amp we chose for this design is the TL072IP from TI. There are two reasons we chose this: (1) it came in the kit that we bought and (2) it is an op amp designed to be used as a comparator.

The plot of input vs output for our op amp as a comparator is shown below in figures 6 and 7. It shows that when the filtered voltage is greater than the reference voltage, the output voltage is 5V and when the filtered voltage is less than the reference voltage, the output is -5V.

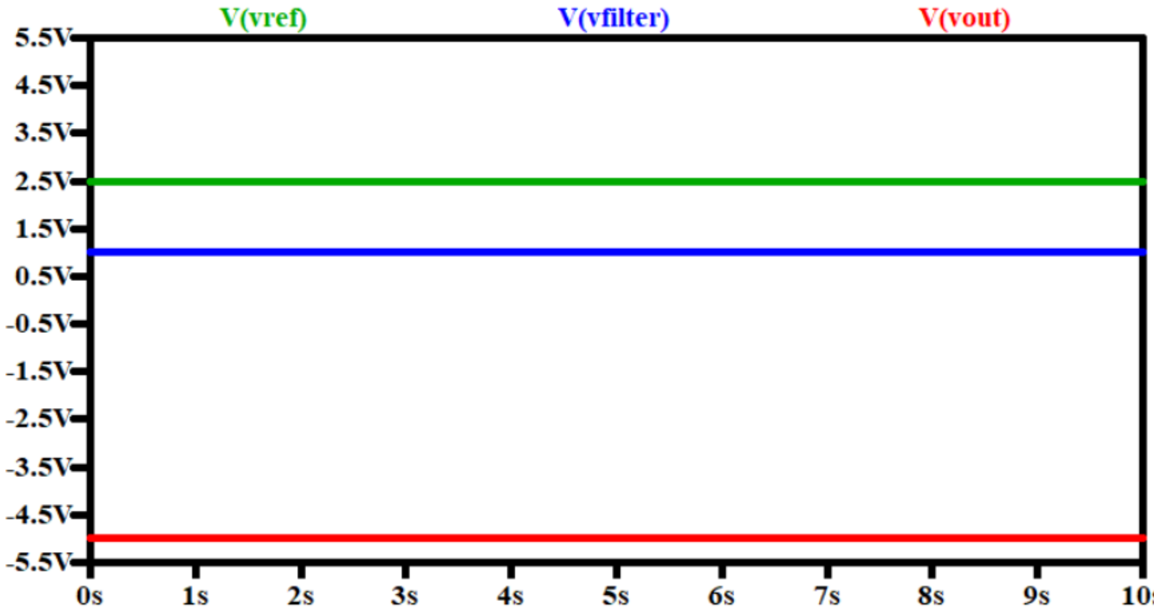


Figure 6: Comparator input vs output when Vref > Vfilter

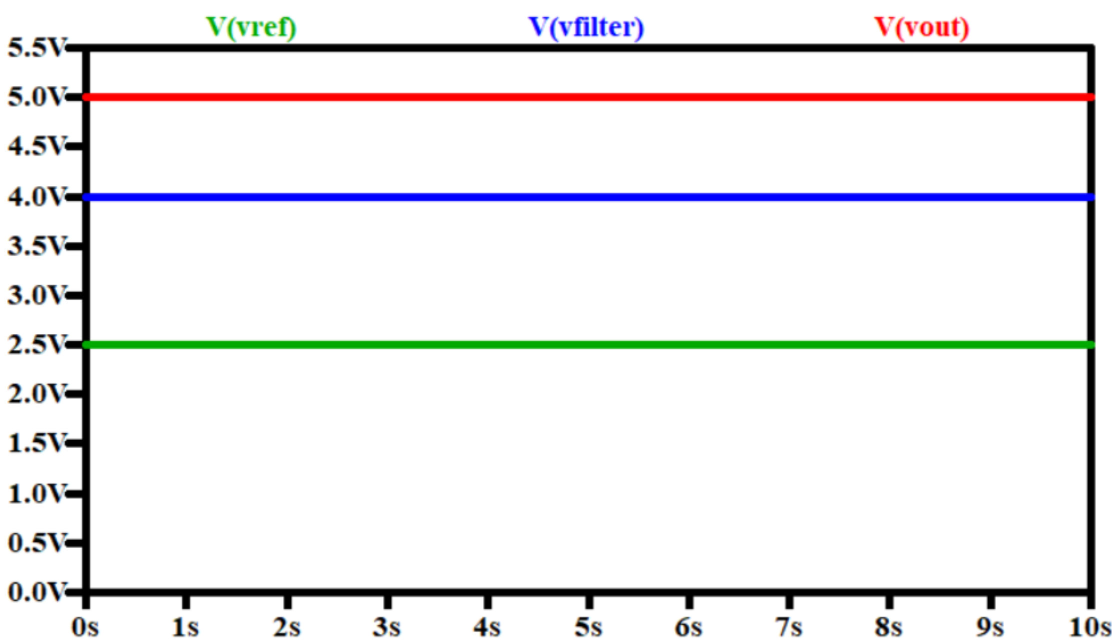


Figure 7: Comparator input vs output when Vfilter > Vref

MS1 Output, LED

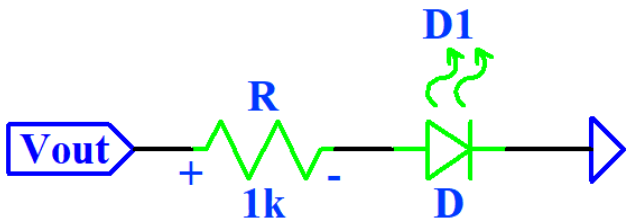


Figure 8: MS 1 Output block schematic

We are using a general 5mm (T-1 ¾) red LED that came in our parts kit. Based on the data sheet, this particular LED has a forward voltage range from 1.7-2.4V [2]. It also has a peak forward current of 100mA [2].

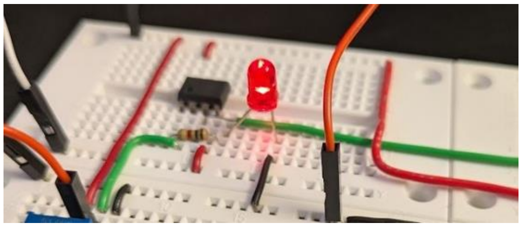
In figure 7, V_{out} is the output voltage from the op amp. So, it will only be 5V or -5V. Below are the two statements for the output:

$$V_{out} = 5V \rightarrow LED\ turns\ on \tag{6}$$

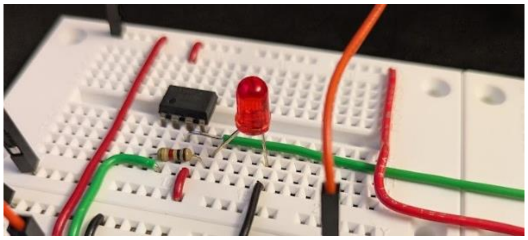
$$V_{out} = -5V \rightarrow LED\ turns\ off \tag{7}$$

To demonstrate this functionality, we took photos of the LED for the cases where V_{out} is +5V and when V_{out} is – 5V.

When our V_{out} is +5V, the LED is on:



And when V_{out} is -5V, the LED is off:



MS2 Building Block, Integrator

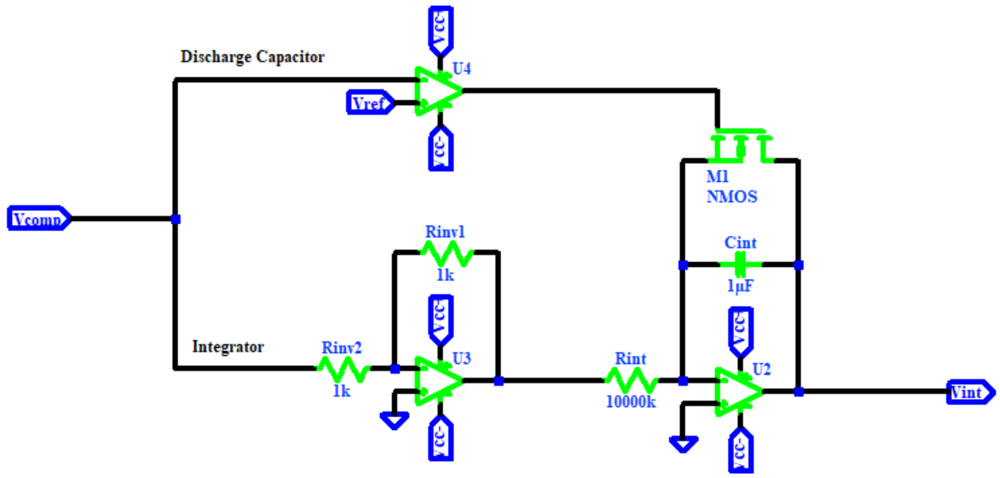


Figure 9: Integrator Schematic

Op amp to discharge the capacitor:

$$V_{ref} > V_{comp} \rightarrow V_{out} = V_{cc+} = 5V \tag{8}$$

$$V_{comp} > V_{ref} \rightarrow V_{out} = V_{cc-} = -5V \tag{9}$$

Since we use a non-amplifying inverter cascaded with an inverting integrator:

$$V_{int} = (-1)(-\frac{1}{R_{int}C_{int}})\int V_{comp}dt \tag{10}$$

To make the integrator ramp in the time of 10s from 0V to 5V, we needed to choose specific values for C_{int} and R_{int}. Using the time constant equation, we found that if we use $C_{int} = 1 * 10^{-6}$, then $R_{int} = 1M\Omega$.

To show the operation of the integrator, the simulation shows how the integrator is triggered by the V_{comp} output from MS1 so when V_{comp} is HIGH, the integrator ramps from 0V to 10V in 10s.

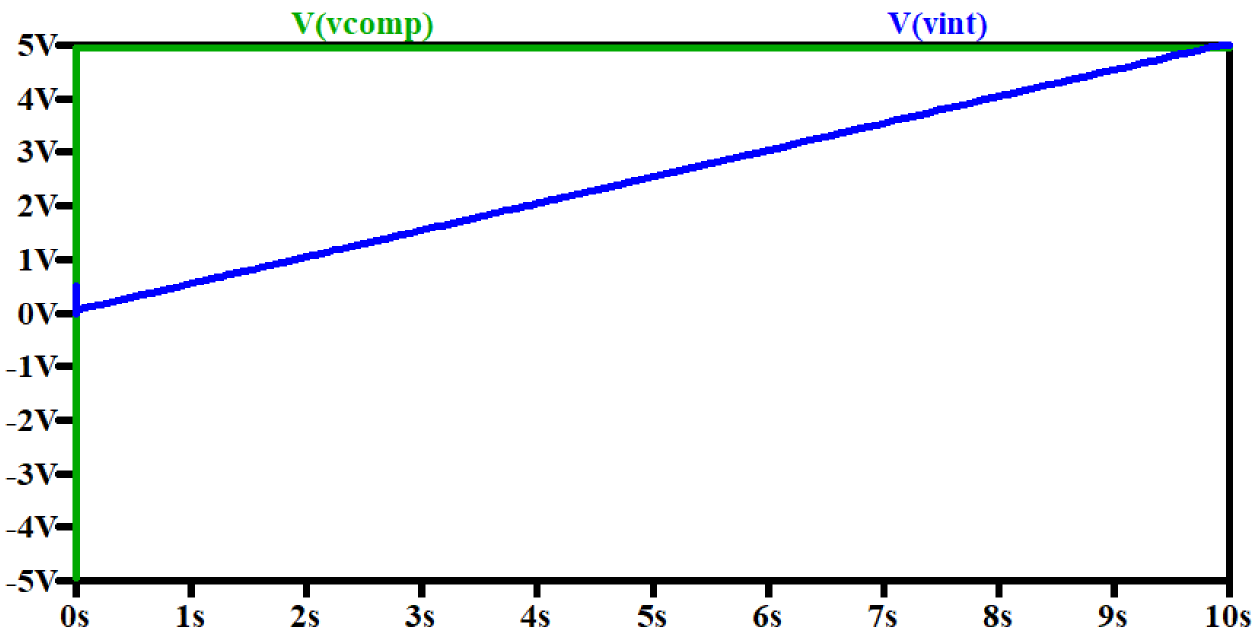


Figure 10: Integrator Output when Vcomp is High

Then when V_{comp} is LOW, the integrator stays at 0V.

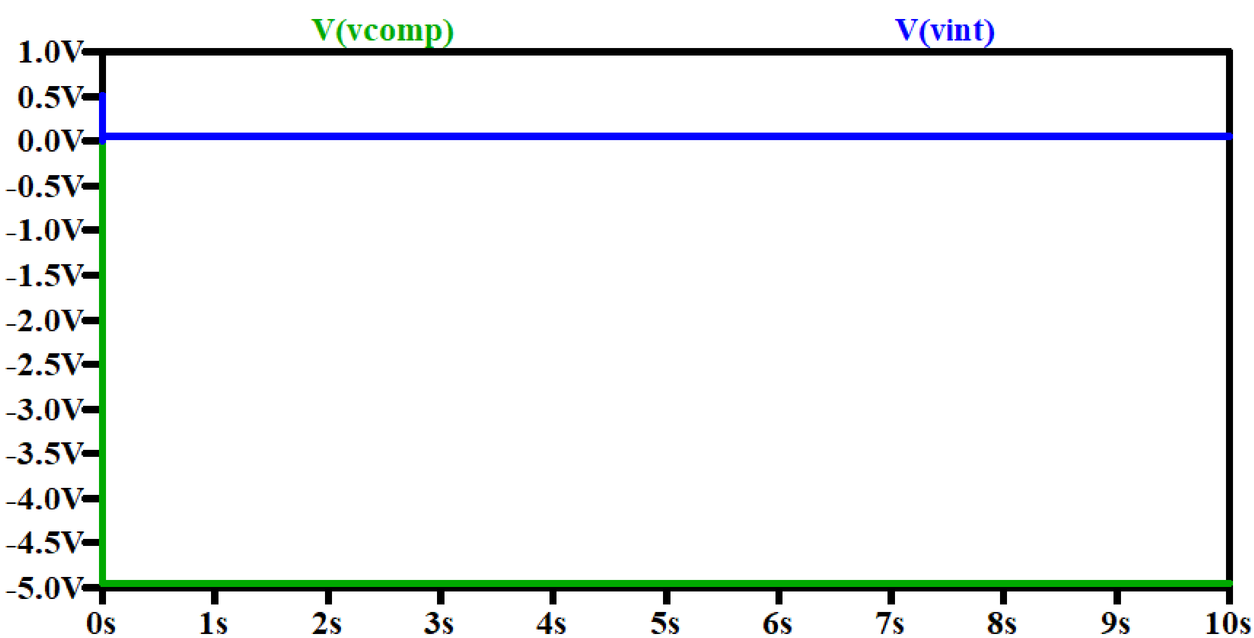


Figure 11: Integrator Output when Vcomp is Low

Also, since we included a system to discharge the capacitor, we can display this by making V_{comp} cycle through HIGH and LOW in the simulation:

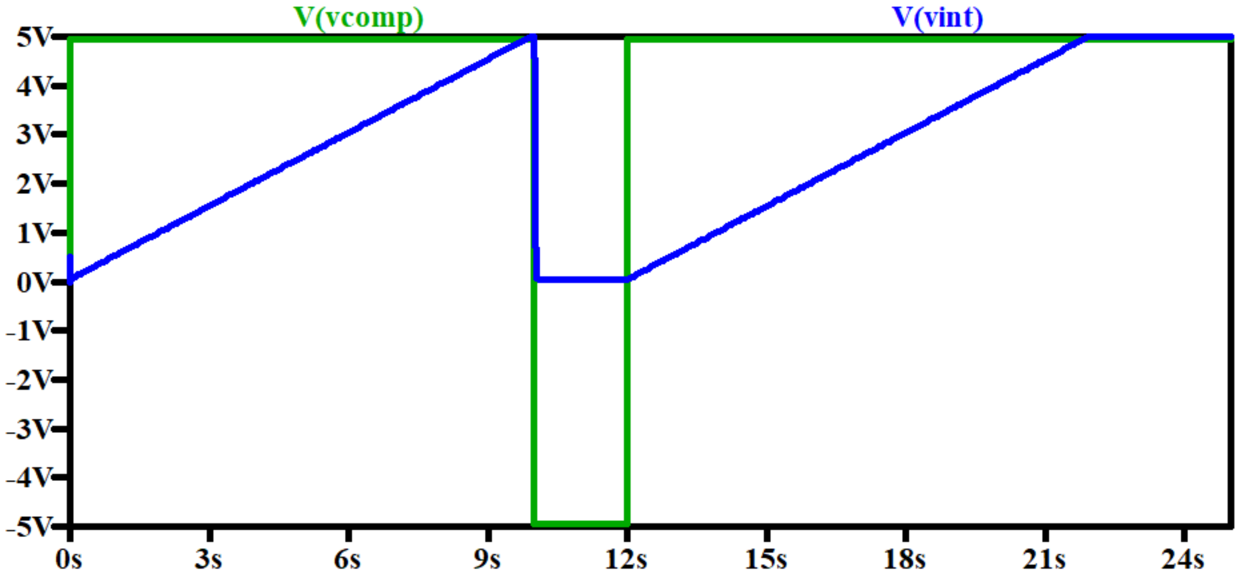


Figure 12: Integrator Output to Demonstrate Capacitor Discharge

So, when $V_{comp}$is HIGH, the integrator is ramping and when V_{comp} becomes LOW, the integrator becomes 0V and when V_{comp} is HIGH again, it starts ramping from 0V to 5V as desired.

MS2 Building Block, Astable Multivibrator

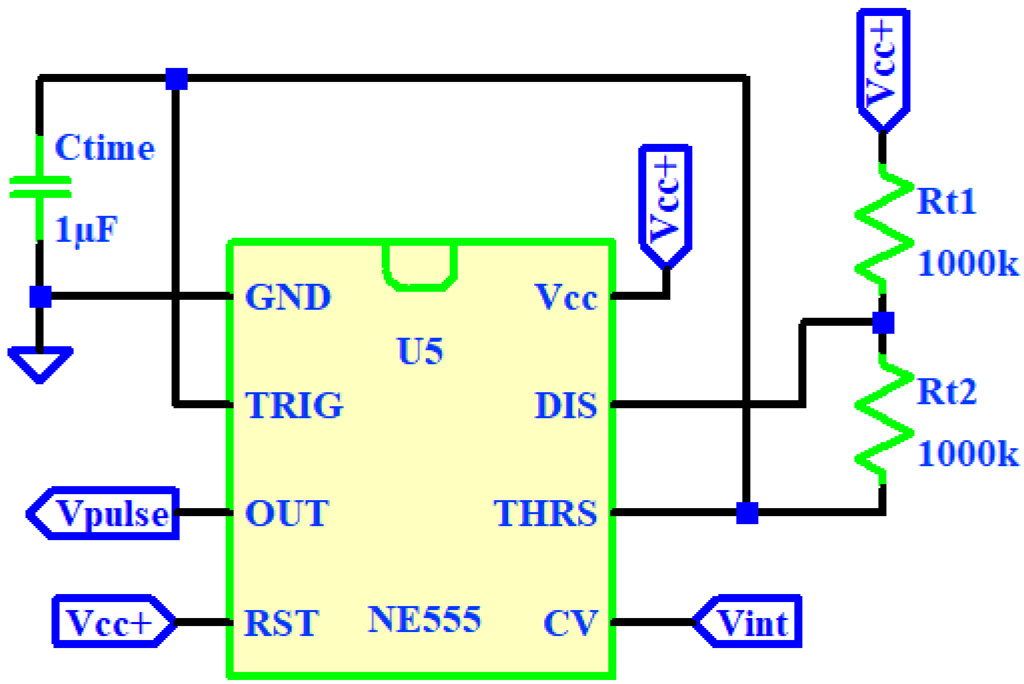


Figure 13: Astable Multivibrator Schematic

For the astable multivibrator, with an input at CV the time low remains the same but time high changes to have an increasing duty cycle:

$$t_{HIGH} = 2\ln\left(\frac{5-0.5V_{int}}{5-V_{int}}\right) \qquad (11)$$

$$t_{LOW} = 0.693(R_{t2})C_{time} \qquad (12)$$

We picked values of $C_{time} = 1 * 10^{-6}F$ and $R_{t1} = R_{t2} = 1M\Omega$ to have a period that is comparable to the 10s that the integrator takes to ramp from 0 to 5V

The action of the astable multivibrator relies on the ramping voltage from the integrator so when the integrator is ramping (V_{comp} is HIGH) then it outputs a pulse voltage with increasing duty cycle:

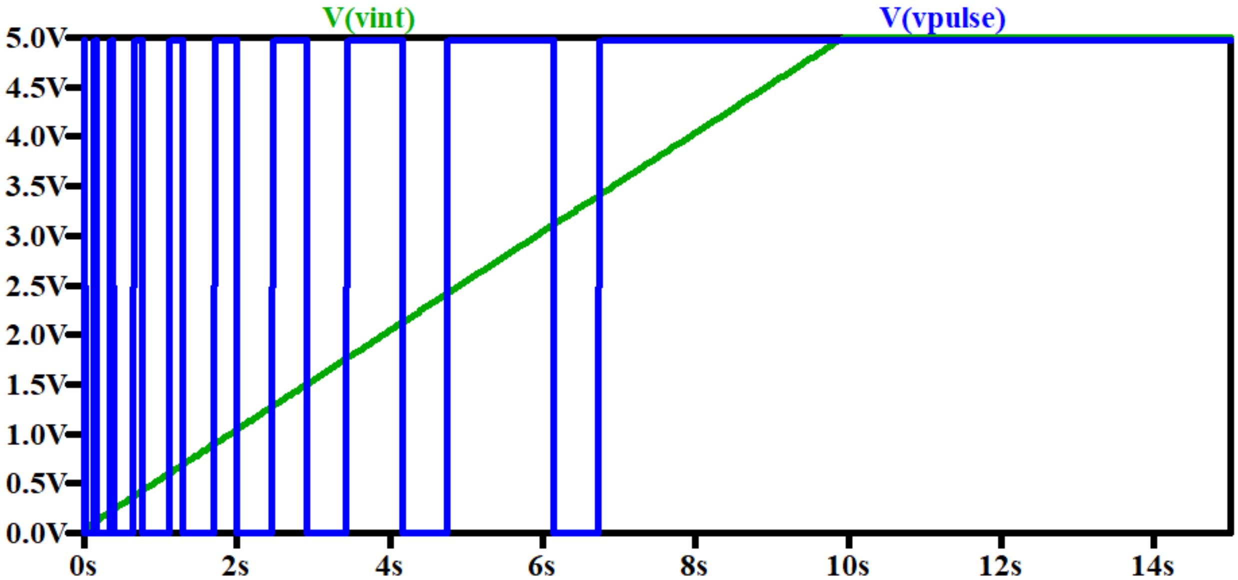


Figure 14: Astable Multivibrator Output When Receiving a Ramping Voltage

Then when the integrator is sitting at 0V ($V_{comp}$is LOW), the pulse output of the astable multivibrator is 0V.

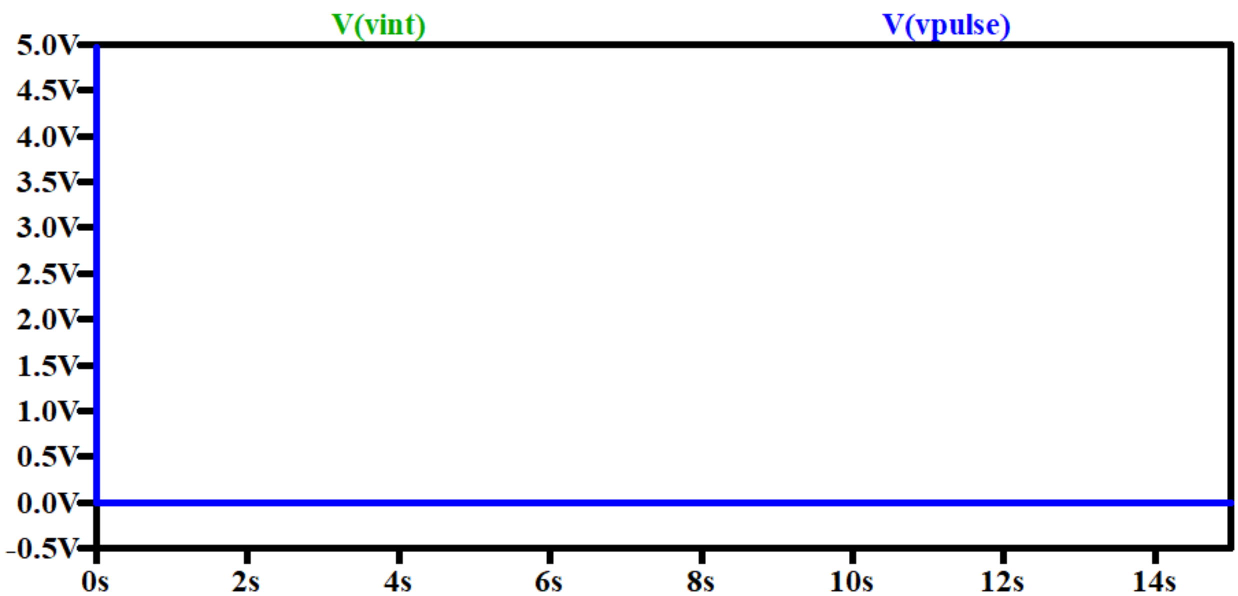


Figure 15: Astable Multivibrator Output When Receiving 0V

MS2 Output, DC Motor

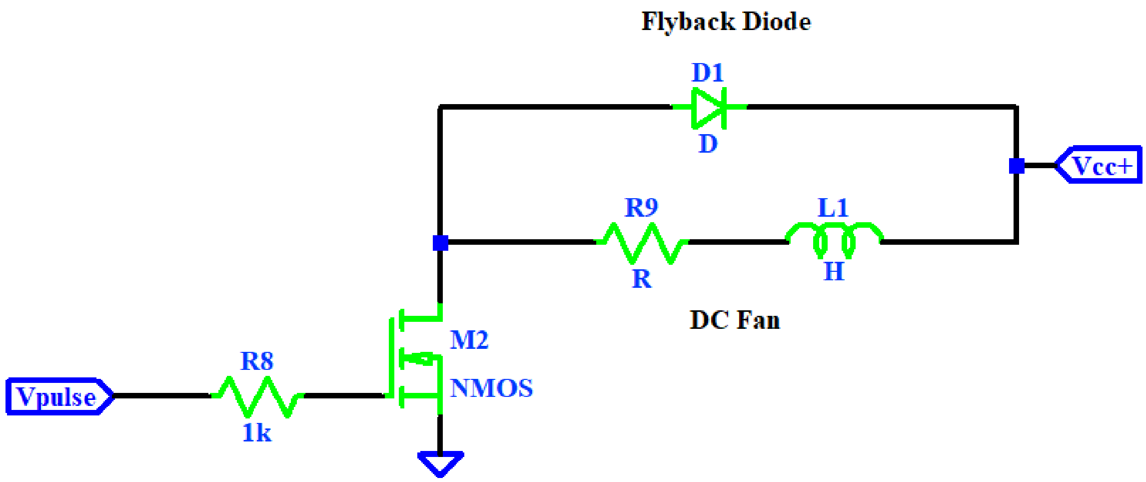


Figure 16: MS2 Output Block Schematic

The DC fan directly reflects the input of V_{pulse} so when V_{pulse} is triggered and outputs a pulse signal, the DC fan is on, otherwise when V_{pulse} is staying at 0V, the fan is off.

Since a DC motor is an inductor, we needed to use a flyback diode in parallel to the motor to protect the transistor from any backwards current that would be generated from the motor after turning off the pulse signal. Also, since the M2K board doesn't have enough current from its voltage sources, we used a +5V source from a microcontroller for the ${V_{CC}}^{+}$ connection to the fan which had over 0.2A.

MS3 Building Block, Oscillator Circuit

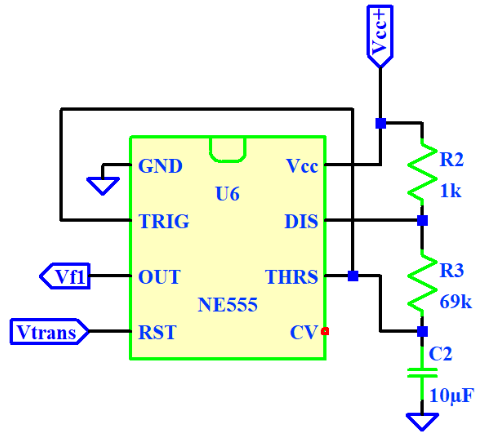


Figure 17: MS3 1Hz Oscillator Circuit

For this oscillator circuit, the only input is V_{trans} which determines if it is on or off, then verify an output of a 1Hz frequency pulse signal:

$$t_{HIGH} = \ln(2) * (R_2 + R_3)(C_2) = 0.485\ s \tag{13}$$

$$t_{LOW} = \ln(2) * (R_3)(C_2) = 0.478\ s \tag{14}$$

$$f_1 = \frac{1}{t_{high} + t_{low}} = 1.038\ Hz \tag{15}$$

For this design, the resistor values were picked as a result of wanting a 1Hz frequency. First for the capacitor, since we are designing for a goal frequency of 1Hz which is relatively slow, C_2 or the resistors need to be relatively large in order to have a larger time constant. In this case, the capacitor was chosen to be a larger value so $C_2 = 10\mu F$. To make the square wave t_{HIGH} approximately equal to the t_{LOW}, the charge resistance and discharge resistance needed to be close and so $R_2 = 1k\Omega$ and $R_3 = 69k\Omega$.

To test this design, simulate on LTSpice with the 555 Timer turned on to get:

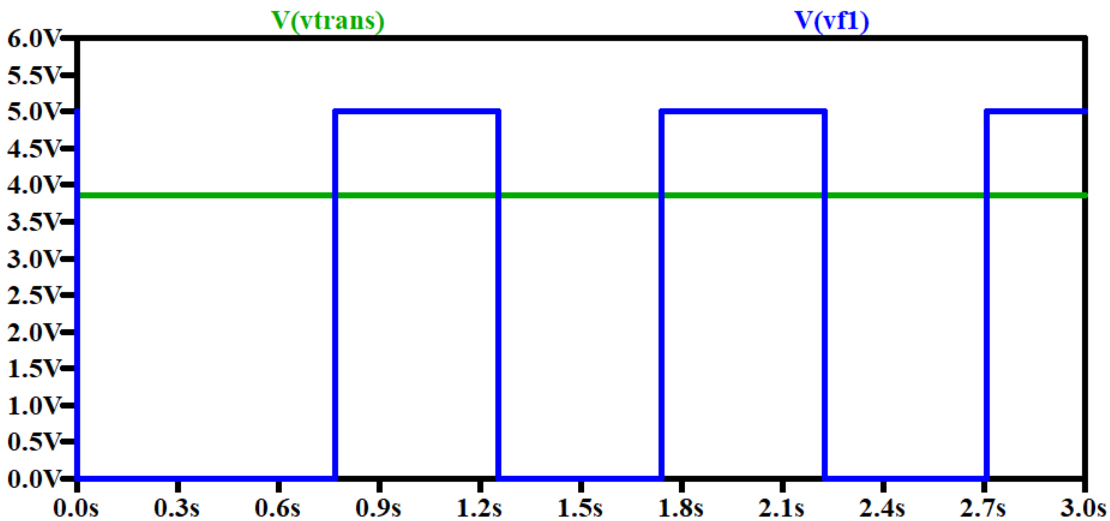


Figure 18: 1Hz Oscillator Output with a 4.0V Input (Logic High)

Where V_{f_1} is the output of the oscillator circuit with an approx frequency of 1.03Hz.

MS3 Building Block, First-Order Filter (Passive, High Pass)

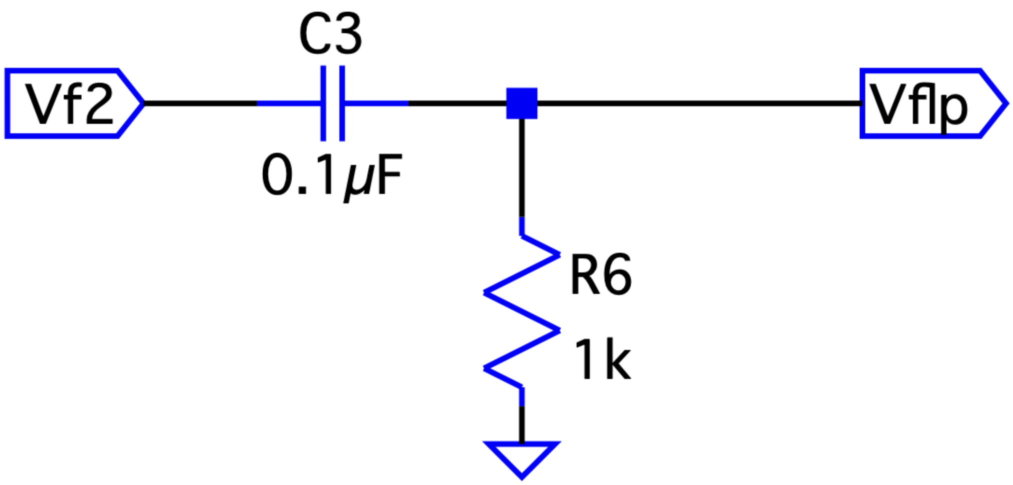


Figure 19: First-Order Filter Schematic

The first order high pass filter filters out signals below a certain angular frequency known as the corner frequency. As there are no op amps involved, this filter is a passive one. It is defined by the following equations:

$$\frac{V_{flp}}{V_{f2}} = \frac{s}{s + \omega_c} \tag{16}$$

where corner frequency ω_c:

$$\omega_c = \frac{1}{R_6 C_3} = \frac{1}{2\pi(0.1 \cdot 10^{-6})(1000)} = 1.59\ kHz \tag{17}$$

The gain is defined by:

$$|H(j\omega)| = \frac{\omega}{\sqrt{\omega^2 + \omega_c^2}} = \frac{\omega}{\sqrt{\omega^2 + 1591^2}} \tag{18}$$

with our output in decibels defined as $20 \log(|H(j\omega)|)$. Gain increases at a rate of 20 db/dec until we reach the corner frequency.

The phase shift is defined by:

$$\theta = \tan^{-1}\left(\frac{\omega}{\omega_c}\right) = \tan^{-1}\left(\frac{\omega}{1591}\right) \tag{19}$$

A simulation of the frequency response of our filter as Vf2 sweeps from 1Hz to 1MHz can be found below. The corner frequency of this particular filter is 1.59kHz.

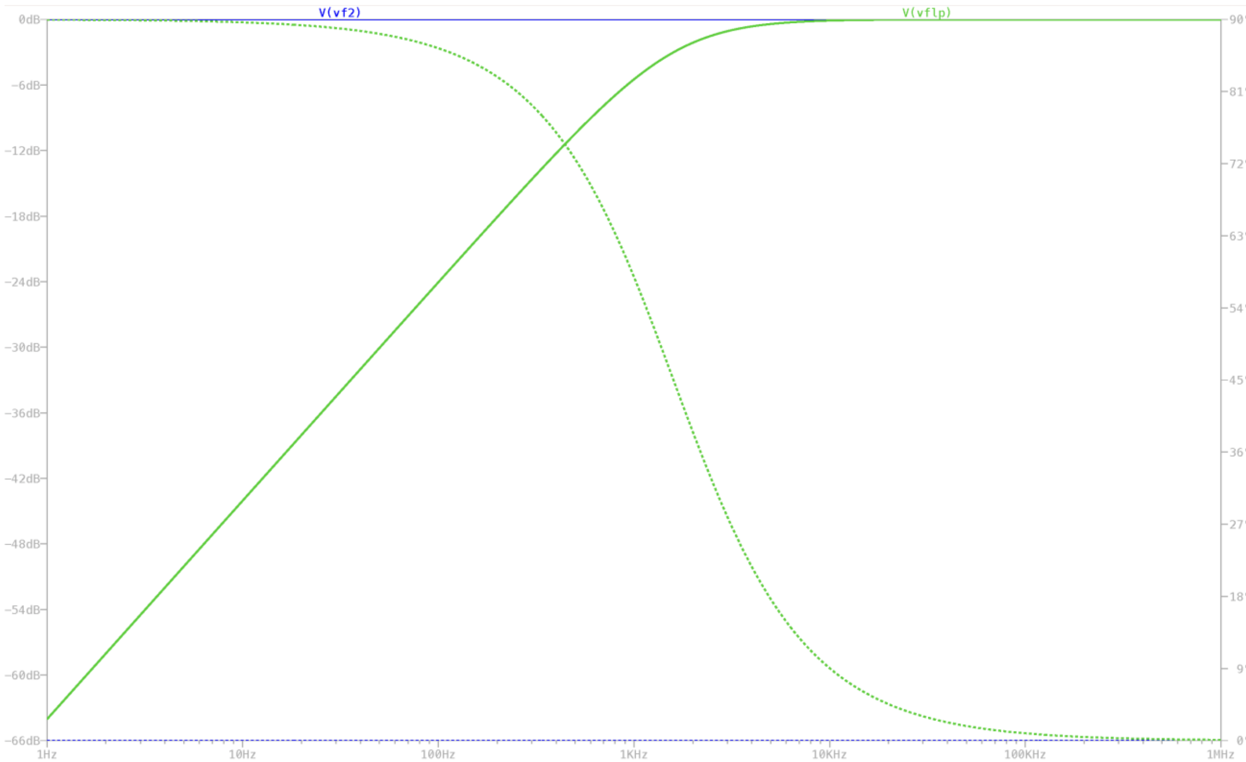


Figure 20: Bode Plot for the First-Order Filter

MS3 Building Block, Second-Order Filter (Active, Low Pass)

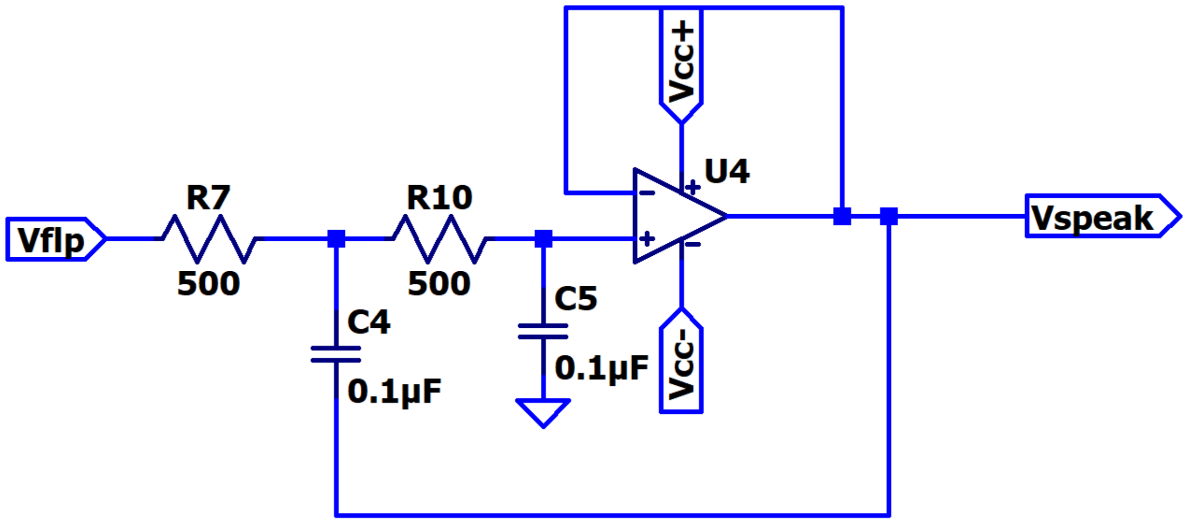


Figure 21: Second-Order Filter Schematic

The transfer function, H(s), is equal to $\frac{V_{out}}{V_{in}}$

$$H(s) = \frac{4 * 10^8}{(s + 2 * 10^4)^2} \tag{20}$$

Since this filter is critically damped, the corner frequency is equal to $2 * 10^4$ rad/s.

$$\frac{2 * 10^4}{2\pi} = 3183Hz = 3.18kHz \tag{21}$$

A simulation of the frequency response of our filter as V_{flp} sweeps from 1Hz to 100kHz can be found below. The corner frequency of this particular filter is 3.18kHz.

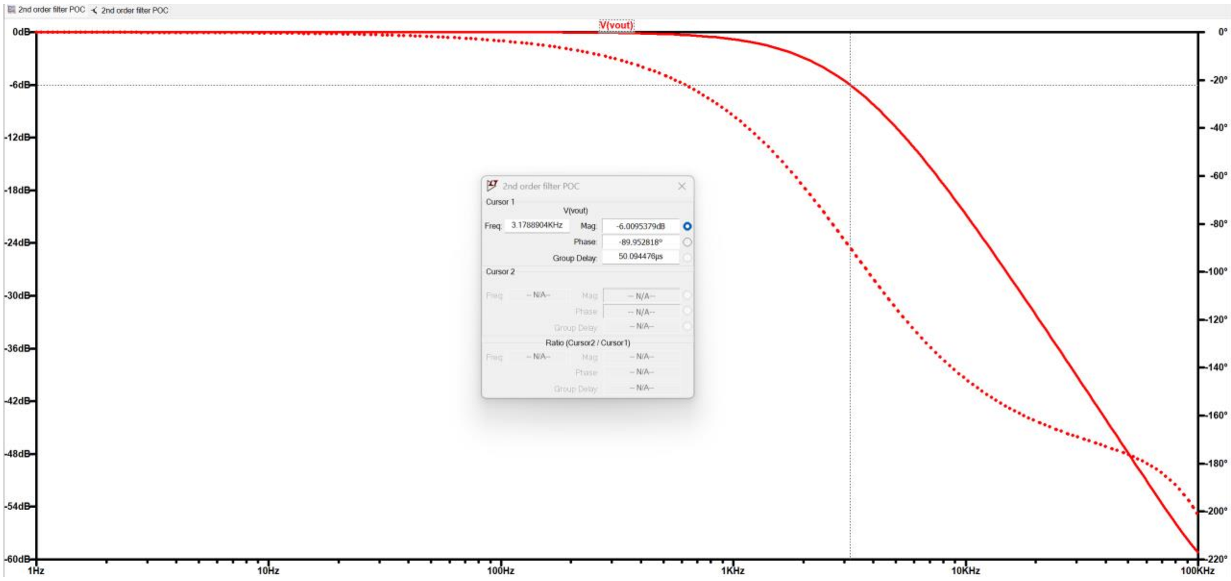


Figure 22: Second-Order Filter Bode Plot

MS3 Output, Speaker

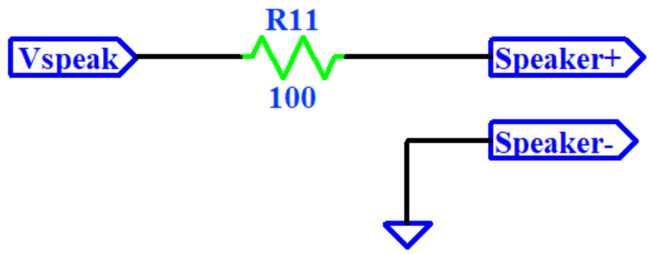


Figure 23: MS3 Output Block (Speaker) Schematic

The speaker output directly takes the output after the 2nd order active low pass filter that we use and will output the signal ideally of around 2kHz to be beeping on an off. This beep will be at a frequency around a 1Hz frequency and be audible for people to hear as it is an alarming system. This speaker output is all initially triggered by using an output from MS1 that uses a comparator to indicate if the levels of CO in an environment are dangerous or not in accordance with OSHA regulations.

Integration and Optimization

The first part of our milestone one is the input block. Since we didn't get the CO sensor in time, the sensor voltage is produced using a 5V source and a potentiometer only for milestone 1. Starting with the sensor voltage, we created a voltage divider using a potentiometer to simulate our CO gas sensor. By adjusting the potentiometer, we can change the two resistor values, causing an output voltage from anywhere between 0 and 5V. We explained this relationship in the building block design equation 3. After this, we designed an RC low pass filter to get rid of possible noise in the sensor voltage output. This filter is low pass thanks to the property of a capacitor which prevents voltage from changing instantaneously like from high frequency signals and can only show change from low frequency signals.

Alongside this input block, we used a potentiometer to set the reference voltage. Since we couldn't use a CO sensor at this stage due to parts not arriving, using a potentiometer would allow us to change the reference voltage to whatever the real reference voltage value will be when 50ppm is detected by the sensor. 50ppm is based on our research on OSHA standards for safe levels of CO in a workplace. In future milestones, we will test the output voltage of the sensor when detecting 50 ppm and set that as our reference voltage. However, for this milestone, we simply chose a reference voltage to be 2.5V while we waited for the part. After deciding on this value, we were able to calculate what the value of the 2 resistors in the voltage divider needed to be. Based on our calculations, R1 had to equal R2 and since we had a 10k ohms potentiometer, each resistor would be set to 5k ohms.

The next building block was the op amp comparator. This was designed to connect our two previous stages together. At this point, we now have a filtered sensor voltage and a reference voltage. Using an op amp, we used it as a comparator to give an different output based on the values of our two voltage inputs. Because we want the light to turn on when the filtered voltage is greater than the reference, we fed the reference voltage to the positive input of the op amp. We then fed the filtered sensor voltage to the negative input of the op amp. Now, using building block design equations 4 and 5 from above, we know when our op amp will output +5V when the filtered sensor voltage is greater than the reference voltage and when the filtered sensor voltage is less than the reference voltage it will output -5V.

To show an output stage, we connected an LED with a current limiting resistor to the comparator output. The LED we used has a forward voltage range from 1.7-2.4V. As stated above, the op amp outputs 5V when the filtered voltage is greater than the reference. This 5V is enough to power our LED. So, to bring it all together, our system is designed so that when the CO sensor (or simulated sensor in milestone 1) detects more than 50 ppm (momentarily 2.5V), it will turn on the LED, indicating there are dangerous levels of CO present.

The overall input and output for milestone one can be found on figures 6 and 7 above. A V_{out} of -5V means that the LED was not on and a V_{out} of 5V means the LED was on.

Our milestone 2 design starts off with the integrator building block. The input for this building block, and thus the entire milestone 2 design, is the output from the op amp comparator from milestone 1. Knowing this, our integrator building block will either be receiving a value of +5V or -5V. Referring to figure 2, this input is connected to a discharge capacitor system as well as an integrator system. If V_{comp} is -5V, meaning there is not dangerous levels of CO, the op amp for the discharge capacitor system will output a positive 5 volts, according to design equation 8. This 5V is connected to NMOS gate. When an NMOS transistor receives a positive voltage, it allows current to flow between the source and the drain. So, since the gate is receiving 5V, current can flow through the transistor which connects the capacitor to ground. This is how we discharge the capacitor. On the other hand, if V_{comp} is +5V, the output of the comparator would be -5V according to design equation 9. This means current will not be able to flow between the source and drain of the transistor. We decided to implement this method of discharging instead of using a large resistor in parallel because it was a much faster method. We wanted our ramping voltage to start at 0 every time no matter what, so the quicker we can discharge the capacitor the better.

Now we will look at the integrating part of this building block. First, the input is brought through a non-amplifying inverter. The reason for this is because our input for the op amp with capacitive feedback is connected to the inverting input. This means if the input into the integrator was 5V, the voltage would steadily decrease from 0 to -5V. This would not give us a PWM output to be used for the fan. So, after implementing this non-amplifying inverter in series with the integrator, we are now able to use design equation 10 to represent the output of the integrator as a function of time. Let's say that there are unsafe levels of CO. The comparator from milestone 1 would output 5V. Using design equation 10 we can calculate the output of the integrator at any time. The output is a linear equation with slope $\frac{5}{10}$ and a y-intercept of 0. This means multiplying the time we want by .5 will give us the output voltage of the integrator at the given time, except the voltage cannot exceed 5V so after it reaches that value, it plateaus until safe CO levels are reached, where it would then very quickly drop back to 0.

The next building block was the astable multivibrator. The function of this building block is to output a PWM like signal to our DC fan. The specific astable multivibrator used in this case was the 555 timer, known for having 3 5k-ohm resistors connected internally that produce a voltage divider network between +Vcc and GND. The resulting output of the non-amplifying inverter is used as the timer's control voltage (CV), which controls the width of the resulting pulse signal and, as such, how much power should go into the resulting fan. Design equations 11 and 12 determine the duty cycle of the circuit, with equation 11 calculating how long the pulse signal remains in a HIGH state and Equation 12 doing the same for a LOW state. Since the output of the non-amplifying inverter is a ramping voltage when CO levels are unsafe, the 555 timer will output a pulse signal with a duty cycle that increases over time until the CV reaches saturation. The pulse signal duty cycle reaches 50% when the control voltage is approximately 2.2651V, with a 66.7% duty cycle being achieved at around 3.333V. The resistors and values were determined so as to have a period most comparable to the 10 seconds it takes for the output of the integrator to reach saturation, as values too large or too small for one of them makes the frequency too fast or too slow to compare.

The final building block of MS2 was the output block, which contains our DC motor, a flyback diode, and a second NMOS transistor. The output of the astable multivibrator is connected to the NMOS gate connected to ground. As such, any positive voltage that the transistor receives will allow current to flow between the source and the drain and allow the circuit to conduct. This makes sure the fan only turns on when the pulse signal is in a HIGH state, so as to not waste energy. Since the fan is an inductor, however, a lingering backwards current can form from repeated flyback caused by the pulse signal. To eliminate this current, a diode is placed with reverse polarity to the power supply – the pulse signal in this case - and parallel to the inductor. When there's no power coming from the astable multivibrator, the diode reroutes forward any lingering current captured by the inductor through it thanks to its minimal resistance.

Milestone 3 begins with the output from the comparator in milestone 1 being connected to the gate of a transistor. If it receives a -5V from the comparator, V_{trans} will be 0V. If it receives +5V, V_{trans} will be 5V.

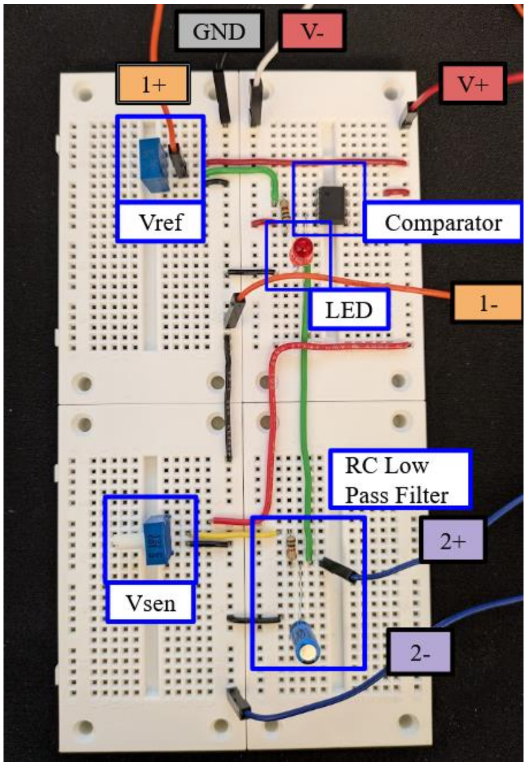
Our first building block is an oscillator circuit and V_{trans} controls when this circuit is active or not. V_{trans} is connected to the active low reset input of the first 555 timer, which acts as our 1Hz oscillator circuit. This means that when V_{comp} is +5V, our oscillator circuit will be active. Using design equations 13 and 14, the output of this oscillator circuit will be high for .485 seconds, low for .478 seconds and then repeat. Then, using design equation 15, we can calculate that the frequency of the output of this first oscillator circuit will be 1.038Hz. This signal is connected to the reset of the 2kHz oscillator circuit. Doing this means that the oscillator circuit that will output the signal received by the speaker will be active for around .5 seconds and then deactivated for around .5 seconds. This allows us to achieve the beeping alarm rather than a long, continuous noise. To get a better understanding, you can see what this output looks like in figure 31 below. Now, using design equations 13 and 14 again with the component values in our 2kHz oscillator circuit we get that the output will be high for $2.98 * 10^{-4}$ seconds, low for $2.29 * 10^{-4}$ seconds, and then repeat while the circuit is not being reset. Using these values in design equation 15, we get a frequency of 1.90kHz. We will call this output V_{f2}.

The next building block is the first-order passive high pass filter. This is a simple RC circuit, and we are using the voltage of the resistor as the output. By doing so, we can find the transfer function of the filter, which is $\frac{V_{flp}}{V_{f2}} = \frac{s}{s+\omega_c}$ (design equation 16). Then, using design equation 17, we can calculate that the cutoff frequency is 1.59 kHz. Since this is a first-order passive high pass filter, there will be a +20 db/decade slope up until the corner frequency. At the corner frequency, the value will be -3dB. At frequencies greater than the corner frequency, the vlaue will be 0dB. This means that the filter passes frequencies greater than 1.59kHz and attenuates the frequencies below that.

After this, the signal is brought to the second-order active low pass filter. For the second-order filter, we are taking voltage at the output of the op amp, and sending this to our speaker. Using design equation 20, we know that the transfer function is $\frac{V_{speak}}{V_{flp}} = \frac{4*10^8}{(s+2*10^4)^2}$. Then, using design equation 21, we calculated the cutoff frequency to be 3.18kHz. Based on the transfer function, we can tell that this filter is critically damped. Knowing that this is a critically damped second order low pass filter, we know that the filter will pass freqeuncies lower than 3.18kHz and attenuate frequencies greater than that. Since its criticall damped, there will be a rolloff of -40db/decade.

Lastly, this filtered signal is brought to our speaker. The will output the signal ideally of around 2kHz to be beeping on and off. This beep will be at a frequency around a 1Hz frequency and be audible for people to hear as it is an alarming system.

Here is some information on our experimental data from MS1:



Each part of our circuit is labeled with blue boxes to indicate what components are in each part.

A guide to the labels:

V- = -5V,

V+ = +5V,

1+/- is measurement channel 1 on the oscilloscope,

2+/- is measurement channel 2 on the oscilloscope,

GND is ground

Figure 24: Milestone 1 Design

Channel 1 is measuring the reference voltage

Channel 2 is measuring the filtered sensor voltage.

When we turn the knob to have $V_{sensor} > V_{ref}$ representing the case where there is more than 50ppm as the CO measurement meaning it is unsafe to work in the condition:

Period:	--	Period:	--
Frequency:	--	Frequency:	--
Peak-peak:	76.256 mV	Peak-peak:	61.005 mV
Mean:	2.490 V	Mean:	3.714 V

The LED turns on:

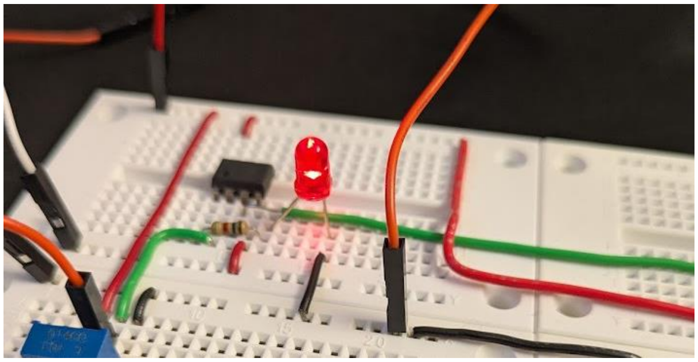


Figure 25: LED when Unsafe Levels of CO

And when we turn the sensor knob the other way to have $V_{sensor} < V_{ref}$ representing the case where there is less than 50ppm as the CO measurement of the room meaning it is safe to work in:

Period:	--	Period:	--
Frequency:	--	Frequency:	--
Peak-peak:	76.256 mV	Peak-peak:	76.256 mV
Mean:	2.498 V	Mean:	1.117 V

The LED turns off:

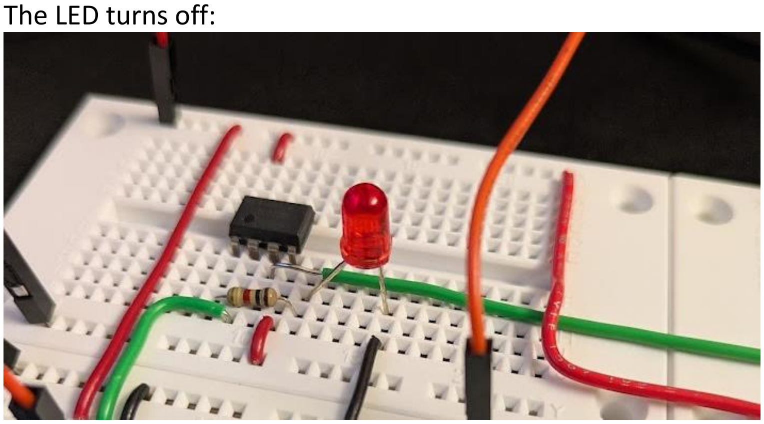
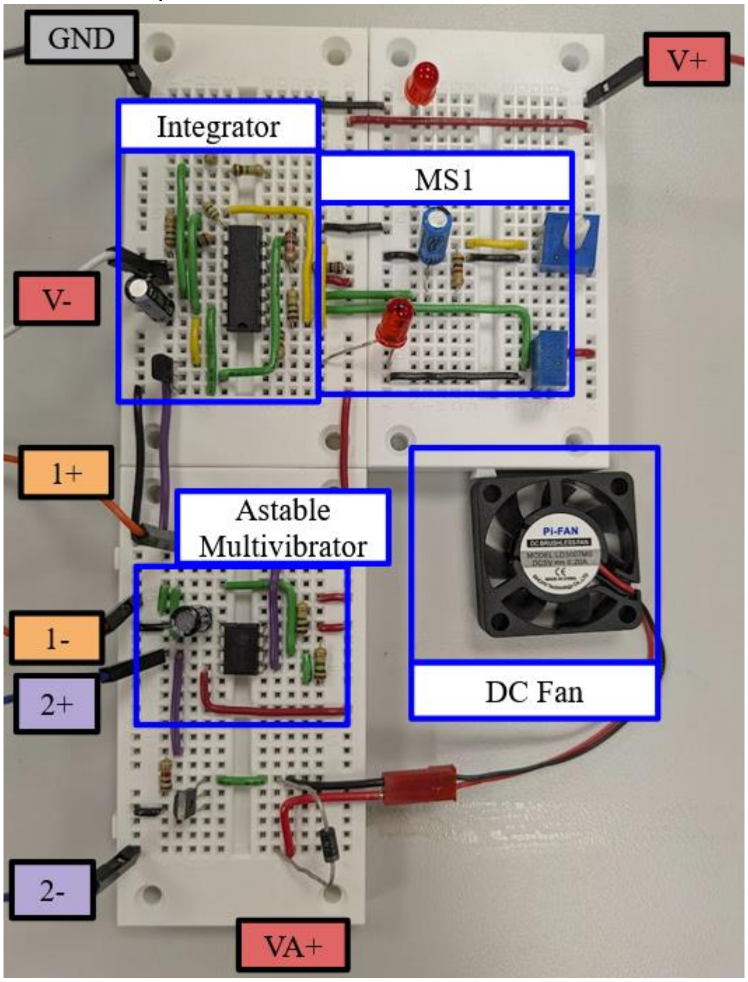


Figure 26: LED when Safe Levels of CO

And some experimental data from MS2:



At the top right is MS1 which is described before this page,

The other boxes represent the blocks of MS2.

V+ = +5V
V- = -5V
VA+ = +5V from Arduino (for current to power DC fan)
GND is ground

1+/- measures the ramping voltage output of the integrator

2+/- measures the pulse output of the astable multivibrator

Figure 27: Milestone 2 Design

When having $V_{sensor} < V_{ref}$ so that V_{comp} is LOW, both V_{int} and V_{pulse} stay at 0V:

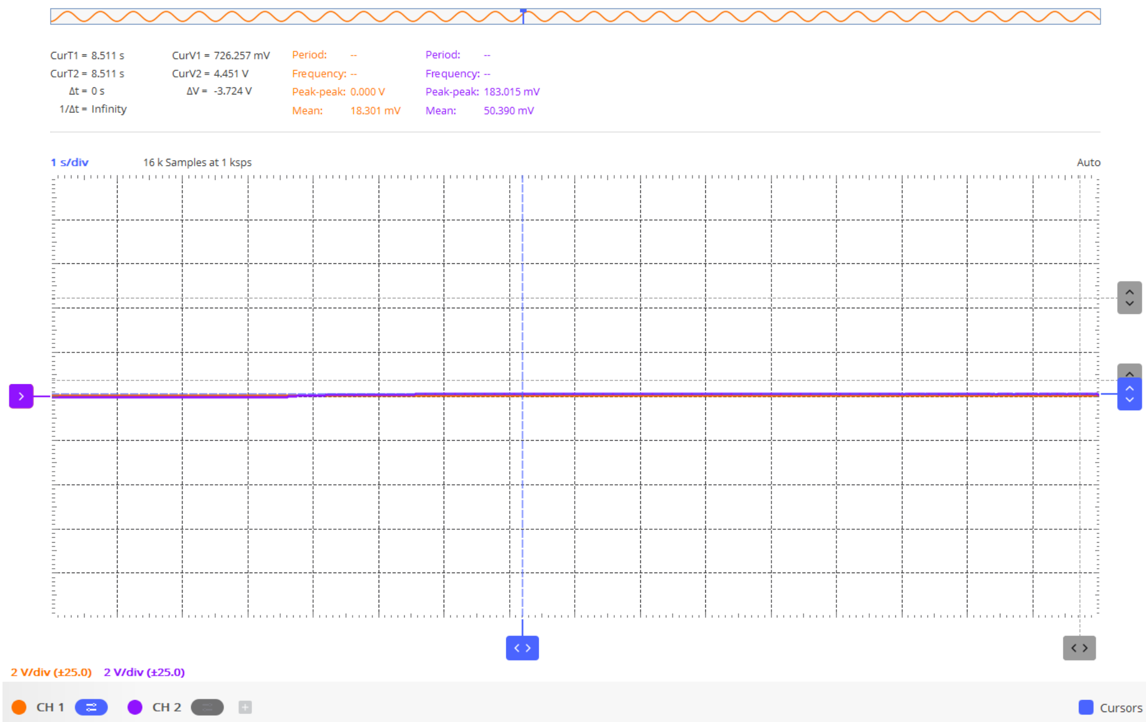


Figure 28: Integrator and Astable Multivibrator Output when Safe Levels of CO

Then when $V_{sensor} > V_{ref}$ so that V_{comp} is HIGH, then V_{int} is ramping up and V_{pulse} is increasing in duty cycle:

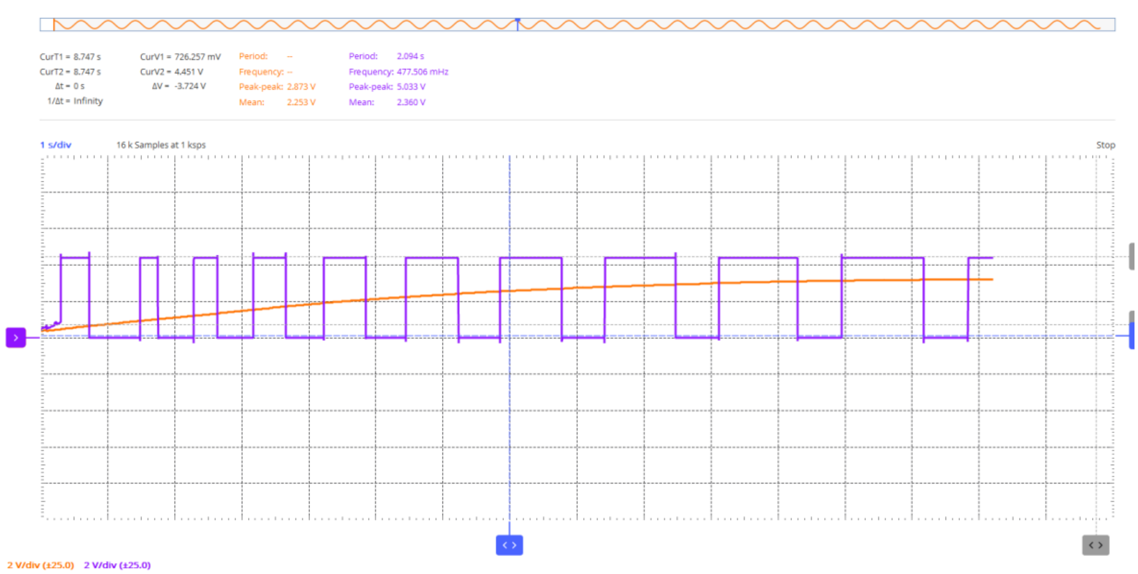


Figure 29: Integrator and Astable Multivibrator Output when Unsafe Levels of CO

Some experimental data from MS3:

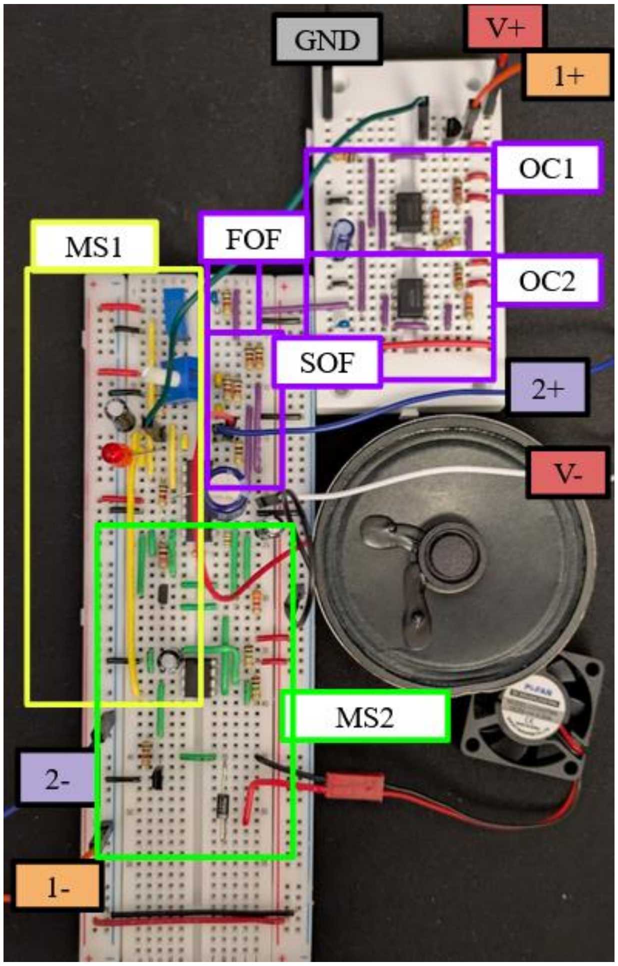


Figure 30: Final Design

V+ = +5V

V- = -5V

GND is ground

MS1 is boxed in yellow

MS2 is boxed in green

MS3 is boxed in purple

Channel 1 measures the comparator output

Channel 2 measures the speaker output

When V_{comp} is HIGH, V_{speak} signal output for the speaker to play:

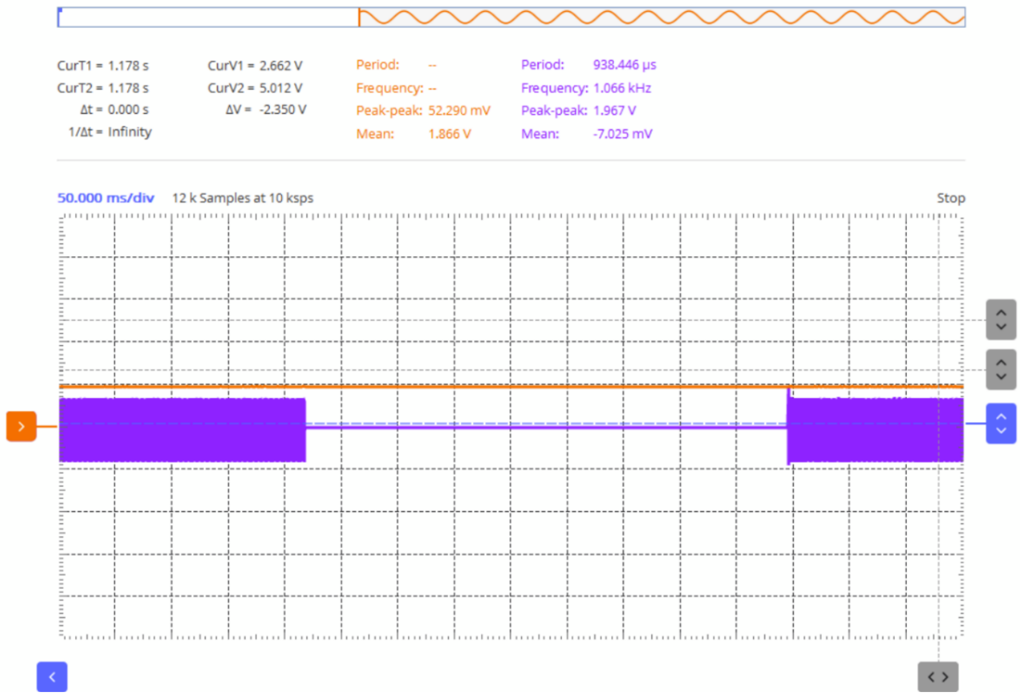


Figure 31: Speaker Output when Unsafe Levels of CO

And when V_{comp} is LOW, V_{speak} is a flat line so no audio will play:

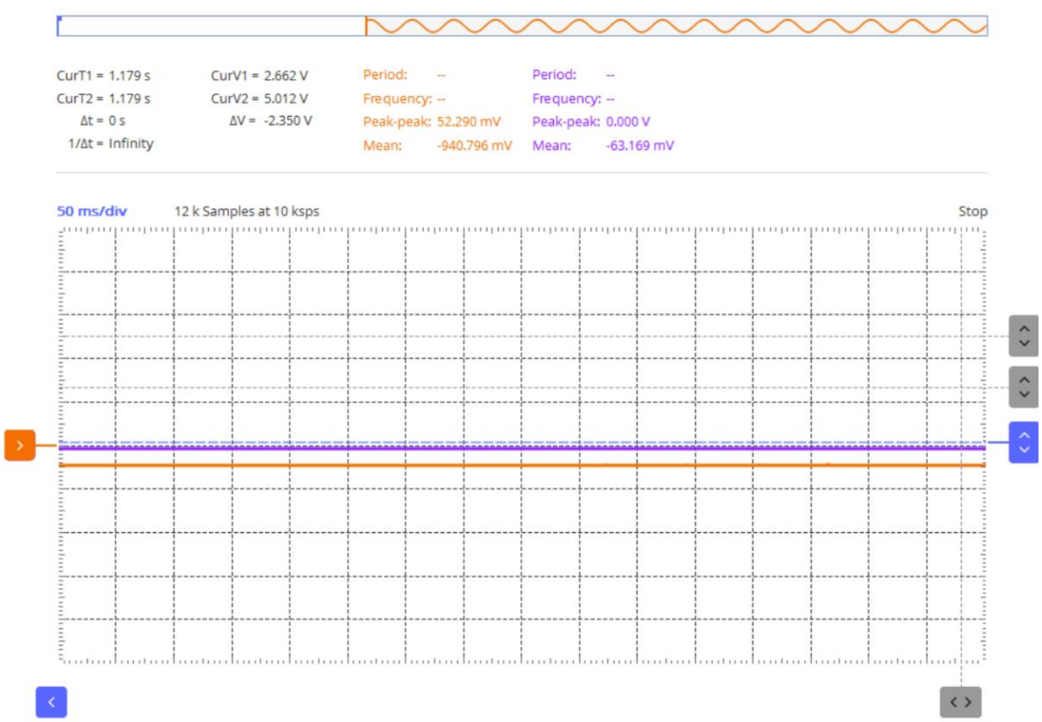


Figure 32: Speaker Output when Safe Levels of CO

Operating Conditions

In MS1, one limitation in our circuit comes from the output voltage of our comparator. Since the op-amp we are using isn't rail-to-rail, it won't reach the saturation values. While it wasn't a big problem in MS1 since the output just relied on either a positive or negative voltage, now that we are using specific voltage values in MS2, we had to switch out the op amp from the TL072IP to OP484 which is rail-to-rail.

Another limitation is the temperature/humidity characteristics of our MQ-7 CO sensor. Based on the data sheet, different environmental conditions can cause the properties of our sensor to shift. The manufacturer did testing from –20 to 50 degrees Celsius. So, this means that in different conditions, our reference sensor voltage will have to be different to achieve the 50 ppm that we want. This might become a problem as it continues to get colder and the only way we have of getting CO right now is from our car. We will try to think of another method to get CO to try and prevent this issue.

One tradeoff is the sensitivity of our circuit for some stability. When we implement our CO sensor, we will set the reference voltage for 50 ppm. If the CO sensor is hovering around that range, even small noise could cause the voltage to continuously cross and recross the reference value. This would show up as a noisy comparator output as well as a flickering LED. To prevent this, we could use a schmitt trigger but the tradeoff with this is that if the gap is too wide, the led (or ventilation system in future milestones) would stay on longer than needed. This would waste energy and our whole goal with milestone 2 is to conserve energy by the ventilation only being on when necessary.

One thing we could do to improve our circuit is to possibly use a different capacitor value in our RC circuit. We could weigh our priorities and choose a capacitor value based on that. If we want quicker detection time and are okay with increasing the risk of flickering caused by noise, we could use a lower capacitor value. If we want to prioritize preventing flickering with the cost of slightly delaying the response, we could use a higher capacitor value.

In MS2, one limitation in our circuit comes from tolerances of the RC components. Since we are using such a large resistor in our integrator, the tolerance can drastically affect the value. This can really affect the slope of our ramp. Luckily, right now we don't need the slope to be an exact value, but it is something to consider.

Another limitation has to do with noise from the comparator in MS1. If the output of the comparator contains even just small amounts of noise, the integrator will integrate that error over time, causing the error to be much more exaggerated than it actually is. This will end up effecting the output of the integrator, and in turn, the rest of milestone 2.

One tradeoff is we decided to use a ramping voltage from an integrator instead of using a PID control system in order to save resources and lower costs. The reason we chose this tradeoff is because the levels of CO don't fluctuate rapidly, it's more of a slow steady change. Because of this, we figured that a PID control wouldn't be as important as it could be in other situations. By choosing this tradeoff, we sacrifice some proportional control, as well as a little bit of predictions, but in this case where levels change slowly, it's a tradeoff we were willing to make.

One thing we noticed during testing and during the presentation is that at lower duty cycles, instead of just spinning slower, the fan would just spin and then stop and repeat. One way we could improve our circuit is by changing this. One thing we could do is set a minimum duty cycle for the fan. For example, if we notice that this problems stops happening once we reach 30% duty cycle, we could implement a way so that the fan only starts spinning at 30% duty cycle.

In milestone 3, one limitation is that our first order filter is only +20db/decade. This is not a very steep roll off which means that frequencies slightly below the corner frequency won't be as attenuated as we would like. Since we are using a speaker to output a tone based on the frequency, this isn't the end of the world, but it is still a limitation.

Another limitation that we found in this milestone was that when the dc fan would turn on, it would pull too much voltage and destabilize the sources. This was a huge limitation as it was messing up all our values, and we weren't getting anything that we expected. To combat this limitation, we added a 100mF capacitor between the Vcc+ and Vcc- of our op484 op amp to act as a "reservoir".

One tradeoff in this milestone was the second order filter choice. We ended up using a critically damped second order filter. By using a critically damped filter, we were able to get a steep roll off of -40 db/decade right at the corner frequency. The tradeoff for this is that the passband isn't quite as flat as it would be for an overdamped filter. We chose this tradeoff because it seemed worth it to have a steeper roll off even if it meant that some frequencies slightly lower than the cutoff frequency would be attenuated a little more than if it were over damped.

One way that our circuit could be improved is that we could implement multiple sensors and alarms. For example, in a building, we could have a sensor in each room. If a sensor detects unsafe levels, all of the alarms in the building will go off, letting everyone know that they should leave the building until it is safe.

Another way that we could improve our circuit is that we could add different thresholds for the amount of CO. For example, at 50 ppm, the ventilation would turn on. If sensors detect 100 ppm, the ventilation will turn on, and the alarm would beep every with a .5Hz frequency. If sensors detect 200 ppm, the ventilation will turn on, and the alarm would beep at a frequency of 2Hz.

Engineering Design Considerations

One of our design considerations was around our reference voltage being used to compare with the CO sensor. We did some research on the threshold for CO levels in workplaces and found information from OSHA that said 50ppm was the threshold value for when CO levels become unsafe in the workplace and violate health and safety regulations. Another design consideration we actually based our milestone 2 objective on was saving on electricity and money used for HVAC ventilation systems. To prevent wasted electricity and therefore higher costs of operation, we created a system that controls both the on/off condition of the fan and the fan speed itself based on how long the sensor is on.

References

[1] *Permissible exposure limits – osha annotated table Z-1*. Occupational Safety and Health Administration. (n.d.). https://www.osha.gov/annotated-pels/table-z-1

[2] Farnell, *"5 mm (T-1 3/4) Round LED – Red Datasheet,"* Farnell, 2021. [Online]. Available: https://www.farnell.com/datasheets/3186871.pdf. [Accessed: 12-Oct-2025].

[3] First Alert, *"CO400 Battery Operated Carbon Monoxide Alarm – Product Information,"* First Alert, 2024. [Online]. Available: https://www.firstalert.com/us/en/products/alarms/carbon-monoxide-alarms/co400-battery-operated-carbon-monoxide-alarm-co400/. [Accessed: 12-Oct-2025].

[4] Olimex Ltd., *SNS-MQ7 Carbon Monoxide Sensor Module Datasheet*, Rev. 1.0, Olimex Ltd., 2013. [Online]. Available: https://www.olimex.com/Products/Components/Sensors/Gas/SNS-MQ7/resources/SNS-MQ7.pdf. [Accessed: 15-Oct-2025].

Appendix

Second order filter transfer function derivation:

Using the ideal op amp equations, we know the voltage at the positive input is equal to the voltage at the negative input. In this filter, our negative input is connected to V_{out}, so $V_2 = V_{out}$.

KCL at node V_{out} using impedances:

$$\frac{V_{out} - V_1}{500} + \frac{sV_{out}}{10^7} = 0$$

$$V_{out}\left(\frac{1}{500} + \frac{s}{10^7}\right) - \frac{V_1}{500} = 0$$

KCL at node V_1 using impedances:

$$\frac{V_1 - V_{in}}{500} + \frac{s(V_1 - V_{out})}{10^7} + \frac{V_1 - V_{out}}{500} = 0$$

$$-V_{out}\left(\frac{s}{10^7} + \frac{1}{500}\right) + V_1\left(\frac{1}{250} + \frac{s}{10^7}\right) = \frac{V_{in}}{500}$$

Then treating V_{in} as a constant we solved for V_{out} using matrices.

$$V_{out} = \frac{4 * 10^8 * V_{in}}{(s + 2 * 10^4)^2}$$

The transfer function, H(s), is equal to $\frac{V_{out}}{V_{in}}$

$$H(s) = \frac{4 * 10^8}{(s + 2 * 10^4)^2}$$